

Reference Books

7. Ebert and Adams, Production/Operations Management, Prentice Hall of India, New Delhi (2007).
8. Chase, R. B., Aquilano, N. J. and Jacob, F. R., Production and Operations Management: manufacturing and services, Tata McGraw Hill, New Delhi (1999).

Evaluation Scheme

S. No.	Evaluation Elements	Weightage (%)
1	MST	30
2	EST	45
3	Sessionals (Assignments/Projects/Quizzes/Seminar presentation)	25

UME418 : Power Plant And Process Utility Systems

L	T	P	Cr
3	0	0	3.0

Course Objective: To impart knowledge on the principle of operation, layouts, components, construction, selection criteria and maintenance and troubleshooting aspects of different types of power plants and industrial utility systems. To impart knowledge on the methods of designing industrial processes and systems using design codes and standards and by developing computer program.

Approved in 113th meeting of the Senate held on September 7, 2024.

Syllabus

Introduction: Energy sources for generation of electric power, types of power plant-their special features and applications, present status and future trends of energy resources, overview of utility systems, project implementation stages, load curves, tariff methods

Conventional Power Generation: site selection, plant layout, steam generators, turbines, fossil and nuclear fuels, pulverizers and coal feeding, mill reject, combustion in furnace, coal handling, ash handling, electrostatic precipitators and bag filters, water systems, condensers, cooling towers, safety aspects, waste disposals, cogeneration, hydroelectric power generation, turbine specific speeds

Non-Conventional Power Generation: Fluidized bed combustion, energy generation through wind, geothermal, tidal and solar energy, IGCC

Process Utility Systems: Bulk solids storage and transport systems – silo/hoppers, conveyors, selection and process and instrumentation diagram for pumps, fans and compressors, piping system design, pipe supports, different valves, fittings, instrumentation and data logging systems, industrial fire protection systems, dust hazards

Assignment (s):

Students in groups of 3 to 4:

- Will design the piping in super-heater and re-heater tubes in boiler and will determine and compare the heat transfer rate at different locations. This is to be done using applicable pressure piping codes (ANSI/ASTM or equivalent).
- Will design an optimized material handling system (coal/ash transport system) by developing a computer program.
- Will select a compressor/pump model for a given duty and prepare the process flow diagram (P&ID).
- Will be introduced to the operation of a pilot plant, use of data logging and instrumentation, analysis of data and process modeling

Course Learning Objectives (CLO)

The students will be able to:

1. Design system/process/components by applying the guidelines of codes, standards and catalogues
2. Develop process flow diagrams (P&IDs)
3. Assess troubleshooting requirements for selected systems, analyze and propose optimum solution
4. Develop process flow models acquire/interpret/analyze data from loggers,